

SAHARA

Stress-Aware Health And Resilience Analytics

AI-Based Predictive Personnel Stress and Welfare Monitoring System for Uniformed Forces

Product Requirements Document | Version 2.0 | Smart India Hackathon Grand Finale

Purpose	SIH Grand Finale build specification, problem statement traceability and judging defence
Primary users	Personnel, Welfare Officers, Counsellors / Medical Officers, Unit Commanders, Roster Planners, Administrators
Deployment posture	Prototype: 100% synthetic data. Production: force controlled private network or approved Government community cloud (MeitY empanelled).
PS ID / Organization	SIH26186 / Ministry of Home Affairs Theme: MedTech / BioTech / HealthTech Category: Software
Team	TribeCoders (Team ID 144977)
Supersedes	PRD v1.0 (SAHARA core loop). v2.0 dated 25 Sep 2026.
Status	Scope frozen for finale build. Tier 1 = must demo, Tier 2 = show, Tier 3 = roadmap only.

One-line pitch

SAHARA predicts acute stress (7-day), emotional fatigue (14-day) and burnout (30-day) risk by comparing each person against their own baseline, asks for only the minimum voluntary evidence it needs, routes explainable alerts to welfare staff (never to discipline), recommends a feasible intervention including roster rebalancing, and proves recovery. All of this runs privacy-first and fully inside the force network.

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1. Executive Summary and Problem Framing

Personnel in CAPFs and the Armed Forces face long deployments, night duty concentration, frequent transfers, separation from family, deferred leave and exposure to traumatic incidents. Welfare today is reactive: stress is noticed through manual observation or self-reporting, often after a crisis. Workload data sits in HRMS, wellbeing information is scattered, and actions are not followed up.

The real problem is not 'detect stress from duty hours'. Strain affects people differently, objective and self-reported signals can disagree, and welfare teams need defensible, minimally invasive evidence before offering support. SAHARA therefore estimates **risk trajectories** for four welfare dimensions, measures its own confidence, requests extra voluntary evidence only when needed, and closes the loop with human-approved intervention and measured recovery.

1.1 Product principles

- **Personal baseline before population threshold.** What is abnormal for this person matters more than a generic cutoff.
- **Multimodal confirmation.** Operational, cognitive, physiological and self-reported evidence are interpreted together.
- **Adaptive evidence acquisition.** When confidence is low, ask for one short optional check instead of over-escalating.
- **Human in the loop.** Welfare staff review every individual alert and own every decision. No automatic action on a person.
- **Welfare-only purpose binding.** Data tagged welfare can never be read by disciplinary, promotion or ACR workflows (technically enforced).
- **Least data necessary.** No microphone, message content, call logs, GPS tracking, face or emotion recognition.
- **Personal benefit first.** Personnel get value (own trend, sleep insight, support) before the organization gets insight.
- **Safety over scoring.** Any crisis signal bypasses the model and goes straight to a human.

2. Problem Statement Traceability Matrix

Every line of the official problem statement maps to a concrete SAHARA feature and a visible proof in the finale demo. Judges can tick this table line by line.

2.1 System capabilities ("The system should")

PS requirement	SAHARA feature	Demo proof
Analyze HR indicators: leave patterns, deployment history, duty schedules, transfer frequency, training commitments, workload trends	HRMS Adapter + Operational Feature Engine (Section 7.1): 26 derived features incl. transfers in 24m, leave fragmentation, workload slope	Feature panel for P-104 shows each indicator vs personal baseline
Optional self-reporting and wellness assessments via secure mobile app	SAHARA Mobile (Android APK + PWA): 10-second check-in, WHO-5 style wellbeing, burnout pulse, device binding, PIN, screenshot block	Live check-in on phone in under 20 seconds
Voluntary biometric and wellness data where authorized	On-device camera PPG, reaction time readiness test, optional Health Connect wearable sleep/HRV. Consent per modality.	Consent toggles; payload shows derived features only

PS requirement	SAHARA feature	Demo proof
Detect behavioral patterns associated with elevated stress	Behavioral Analytics Engine: personal baseline deviation, Isolation Forest anomaly, change-point detection, engagement pattern shifts	Baseline chart with deviation band and change-point marker
Generate risk assessments and welfare recommendations for welfare officers and commanders	4-dimension risk model + Intervention Recommender; welfare officers see individuals (pseudonymous), commanders see aggregates	Welfare queue vs commander dashboard side by side
Enable proactive counseling, welfare interventions and workload balancing	Counseling booking, Welfare Need module, OR-Tools roster optimizer, counterfactual 'what would help'	Redistribute 2 night duties, projected burnout drop shown
Strong privacy safeguards, welfare not disciplinary	Privacy Gateway, purpose-binding ('welfare-only' tag blocks use in disciplinary APIs), My Data Mirror, DP aggregates	Commander tries individual endpoint, gets 403, and it is logged

2.2 Expected solution components

PS component	SAHARA module	Tier
Personnel Wellness Monitoring Dashboard	Welfare Officer Console + Commander Readiness Dashboard + Personnel 'My Trend'	Tier 1
Mobile-based Wellness and Self-Assessment Application	SAHARA Mobile (Android APK/PWA, offline-first, multilingual)	Tier 1
Predictive Behavioral Analytics Engine	Baseline + Anomaly + Change-point + Trajectory engine (Section 8)	Tier 1
Stress and Burnout Risk Prediction Models	Four calibrated models: Acute Stress, Burnout, Emotional Fatigue, Welfare Concern + 14-day unit forecast	Tier 1
Welfare Intervention Recommendation System	Rule + model hybrid recommender, counterfactuals, roster optimizer, effectiveness tracker	Tier 1
Role-based Access Control and Privacy Management Framework	RBAC + ABAC (unit, purpose), consent manager, purpose-binding, break-glass, DPDP rights portal	Tier 1
Automated Alerts for authorized welfare personnel	Tiered Alert Engine with SLA escalation, push/SMS/email, crisis fast lane	Tier 1
Data anonymization and secure storage	Pseudonym vault, k-anonymity, differential privacy, AES-256, hash-chained audit	Tier 1

2.3 Preliminary scope, technical challenges and benefits

PS item	Where addressed
Scope 1-5: analytics algorithms, mobile platform, risk engine, dashboards, recommendation system	Sections 6-12
Scope 6: Secure integration with HRMS	Section 13.3: HRMS Adapter (REST/CSV/SFTP), schema mapping, mTLS service identity, idempotent ingestion
Scope 7: Privacy-preserving analytics and RBAC	Section 14
Challenge 1: Privacy and confidentiality	Privacy Gateway, minimization, on-prem, DP (Section 14)
Challenge 2: Preventing stigmatization	Non-diagnostic language, pseudonymous queue, commander sees no individuals, purpose-binding, no label visible to peers

PS item	Where addressed
Challenge 3: False positives / negatives	Personal baselines, multimodal agreement, calibrated confidence, adaptive evidence request, recall at review capacity metric
Challenge 4: Ethical and transparent AI	SHAP + counterfactuals, model cards, fairness audit, human-in-the-loop, Ethics Board review
Challenge 5: Cyber threats	STRIDE threat model, OWASP ASVS L2 controls, mTLS, HSM keys, SIEM, CERT-In reporting (Section 14.4)
Challenge 6: Building trust	My Data Mirror, consent receipts, withdraw anytime, personal benefit first ('My Trend'), buddy system
Benefits 1-8	Mapped to metrics in Section 18 (early identification lead time, incident trend, retention risk, workload balance index)
Strategic: indigenous, Indian operational/cultural context	Section 15: languages, kiosk/offline for remote posts, Tele-MANAS, DPDP, on-prem Indian stack
Potential market	Section 22

3. What Changed from v1.0

Area	v1.0	v2.0 upgrade	Why (PS link)
HR indicators	Duty, night shift, deployment, leave deferral, training	Added transfer frequency , full leave pattern analytics, workload trend slope, hardship posting index	PS: 'leave patterns ... transfer frequency ... workload trends'
Risk outputs	Single 'welfare concern' band	Four named risk dimensions : Acute Stress, Burnout, Emotional Fatigue, Welfare Concern (non-diagnostic)	PS: 'Stress and Burnout Risk Prediction Models'
Welfare concerns	Psychological signals only	Welfare Need & Grievance module (family, housing, finance, children education, leave, medical of dependants)	PS: '...and welfare concerns'
Alerts	Queue only	Tiered Automated Alert Engine with SLA, escalation and multi-channel notifications	PS: 'Automated Alerts for authorized welfare personnel'
Safety	Suicide risk out of scope, no safety route	Crisis Safe-Route : immediate human + Tele-MANAS 14416 routing, no scoring	PS: 'reduction in stress-related incidents'
Prediction	Individual trajectory only	Unit Welfare Weather Forecast : 14-day forward strain from planned roster/deployment	PS: 'Predictive ... proactive'
Explanation	SHAP top factors	SHAP + Counterfactual 'What Would Help' linked to roster optimizer	PS: 'ethical and transparent AI'
Trust	Consent + audit	My Data Mirror (personnel see who viewed their data and why), buddy system, anonymous counsellor chat	PS: 'Building trust among personnel'
Anonymization	Pseudonyms, n>=10 suppression	Minimum cell size + Differential Privacy on commander aggregates with privacy budget	PS: 'Data anonymization'
Planning	None	Welfare Resource Planner : counsellor demand forecast, aggregate retention risk	PS benefits 6 and 7
Biometrics	Camera PPG	+ optional wearable sleep/HRV via Android Health Connect	PS: 'voluntary biometric ... data'

Area	v1.0	v2.0 upgrade	Why (PS link)
Indian context	Generic	DPDP Act 2023 compliance map, CERT-In, Hindi + regional languages, offline/kiosk mode, Tele-MANAS	PS: 'indigenous capability ... Indian CAPFs'
Hackathon plan	6-week roadmap, 5 roles	Pre-finale plan + 36-hour finale plan, 6 roles , judge round strategy	SIH format
Market	None	Market, scalability and adoption section	PS: 'Potential Market'

4. Users, Personas and Permissions

Persona	Needs	Can see	Cannot see
Personnel member (Jawan / Officer)	Fast respectful check-in, confidential help, control over data, no punishment	Own trend, own risk band in plain language, who accessed their data	Anyone else's data
Buddy (opt-in peer)	Know when a nominated peer wants a call	Only a 'please check on me' nudge that the person triggers	Scores, assessments
Welfare Officer	Prioritized queue, evidence, case notes, follow-ups	Pseudonymous individual evidence for own units	Identity (unless approved reveal), other units
Counsellor / Medical Officer	Referral context, appointment, confidential notes	Referred cases with consent	Commander views, roster data
Unit Commander	Unit readiness, workload hotspots, forecast	DP-protected aggregates (n>=10), roster options	Any individual wellness data or identity
Roster Planner	Feasible duty alternatives	Availability, skills, anonymized strain weights	Assessments, case notes
Administrator / Auditor	Onboarding, policy, audit	Config, audit logs, model registry	Welfare case content

5. Scope

In scope (SIH prototype)	Explicit non-goals
Synthetic HRMS for 300 personnel across 3 battalions, 180 days	Clinical diagnosis, therapy, medication, suicide risk scoring
Mobile app: consent, check-in, wellbeing and burnout pulse, reaction test, camera PPG, Health Connect	Continuous monitoring, microphone, messages, call logs, GPS, face or emotion recognition
4-dimension risk model, baseline, anomaly, forecast, SHAP, counterfactuals	Any disciplinary, promotion, ACR, leave denial or automatic roster change
Welfare console, commander dashboard, alert engine, crisis safe-route	Public cloud LLM processing of personnel data
Roster optimizer, welfare need module, resource planner, My Data Mirror	Claims of validated clinical accuracy on real force data

6. SAHARA Mobile: Wellness and Self-Assessment App

Flow

Consent (per modality) → 10-sec check-in → Optional readiness test → My Trend + tips → Request support / Buddy / Welfare need

Feature	Detail	Tier
Granular consent	Separate toggles for check-in, readiness test, camera PPG, wearable, HRMS linkage for prediction. Consent receipt stored (DPDP).	1
10-second check-in	3 emoji sliders: energy, sleep quality, workload feeling. Icon-based for low literacy.	1
Weekly wellbeing pulse	WHO-5 Wellbeing Index (free to use) + 4-item burnout pulse adapted from the Copenhagen Burnout Inventory (public domain) + emotional fatigue item + one optional item: 'Do you need urgent support right now?'. No licensed instruments (e.g., MBI) used. Non-diagnostic.	1
Digital Readiness Check	90-second touchscreen reaction test (PVT style): median RT, lapses, false starts, computed on device	1
Camera PPG (optional)	30-second fingertip on camera: heart rate + HRV proxy computed on device; frames never leave the phone. Fallback: seeded demo value if device torch unsupported.	1
Wearable connector (optional)	Android Health Connect read of sleep duration, resting HR, HRV summary (daily aggregates only). Needs native Android build (Capacitor plugin); PWA shows simulated feed.	2
My Trend	Personal chart in plain language: 'your sleep is below your usual for 9 days'. Micro-tips (sleep hygiene, breathing).	1
Support request	One tap: talk to welfare officer, counsellor, or anonymous chat. Choice of named or anonymous.	1
Welfare need / grievance	Raise family, housing, finance, leave, child education, dependant medical need; tracked with status	1
Buddy system	Nominate up to 2 trusted peers; 'please check on me' nudge without sharing any data	2
My Data Mirror	Timeline of every access to my data: role, purpose, time. Download or request correction/erasure.	1
Security	Device binding, app PIN/biometric unlock, screenshot block, encrypted local store, auto-wipe after inactivity	1
Languages and access	English, Hindi + 2 regional (e.g., Tamil, Bengali) via i18n; large text; offline-first with sync queue	1 (EN/HI), 2 (others)
Kiosk mode	Shared unit tablet for posts where personal phones are restricted; per-person PIN, session auto-clear	2

7. Signal Domains and Features

7.1 Operational (HRMS) indicators

PS indicator	Derived features (7d / 30d / 90d windows)
Leave patterns	Days since last home leave; leave deferred/cancelled count; sudden short-leave frequency; sick-leave spikes vs baseline; leave fragmentation index; leave-balance hoarding
Deployment history	Continuous deployment days; hardship posting index (terrain, altitude, insurgency tier); cumulative deployments 24m; time since last rotation
Duty schedules	Night duty count and clustering; consecutive duty days; rest gap below policy; shift irregularity (circadian disruption score)
Transfer frequency	Transfers in 24m and 60m; days since last transfer; family separation flag after transfer; distance from home station
Training commitments	Training hours added on top of duty; training during recovery window; course failure/retake count
Workload trends	Duty hours slope (linear trend over 30d); workload vs unit median; overtime; staffing gap of unit (demand/strength)
Critical incident exposure	Count of logged operational incidents attended (casualty, IED, disaster) with decay weighting

7.2 Individual evidence domains

Domain	Features	Collection boundary
Cognitive readiness	Median RT, lapses (>500 ms), variability, false starts, test quality	Computed on phone; summary only
Physiological proxy	Resting HR, HRV (RMSSD proxy), sleep duration and regularity, signal quality	On-device PPG or Health Connect daily aggregates
Self-reported	Energy, sleep, workload, wellbeing score, burnout pulse, emotional fatigue, support request	Voluntary form
Engagement pattern	Check-in regularity change (only if user consented; never penalizing, used only to lower confidence or prompt gently)	Metadata of voluntary use
Welfare need	Open welfare requests, category, age of unresolved need	Welfare Need module
Quality and trajectory	Robust z-score, 7/30-day delta, change-points, missingness, modality age	Feature layer

8. Predictive Behavioral Analytics Engine

8.1 Four risk dimensions (non-diagnostic)

Dimension	Definition in SAHARA	Primary drivers	Horizon
Acute Stress Risk	Short-term elevation of strain vs personal baseline	Recent night clustering, incident exposure, readiness drop, HR/HRV shift, low check-ins	7 days
Burnout Risk	Sustained exhaustion and detachment built over weeks	Workload slope, leave deprivation, deployment length, burnout pulse, sleep debt	30 days
Emotional Fatigue	Wearing down of emotional energy and wellbeing	Wellbeing score trend, family separation, transfer recency, self-report discordance	14 days

Dimension	Definition in SAHARA	Primary drivers	Horizon
Welfare Concern	Unmet practical needs likely to drive strain	Open welfare needs, leave cancelled, transfer far from home, dependant medical	Current

Each dimension outputs a **band** (Low / Moderate / High), a **calibrated probability**, a **confidence** score, **evidence coverage** and **top factors**. The UI never uses words like 'depressed', 'disorder' or 'unfit'.

8.2 Model components

Component	Method	Output
Personal baseline	Robust rolling median and MAD after 21-day window	Per-signal robust z-score
Cold start	Hierarchical shrinkage: blend cohort baseline (rank, duty type, posting tier) with personal data, weight grows with days observed	Usable baseline from day 1
Anomaly detector	Isolation Forest on multivariate deviation vector	Unusual pattern score
Change-point detection	PELT / CUSUM on key signals	Date when trend shifted (shown on chart)
Risk models (x4)	Gradient boosting (XGBoost/LightGBM) per dimension on rolling features	Probability of deterioration within horizon
Calibration	Isotonic regression on validation split	Reliable probabilities (Brier, ECE)
Evidence agreement	Domain votes weighted by quality and recency	Agreement / discordance / missing
Explanation	TreeSHAP + human-readable rule mapping	Top 3 contributors with direction
Counterfactual 'What would help'	Constrained search over actionable features only (night duties, leave, training, rest gap)	'2 fewer night duties + 3-day rest: Burnout High to Moderate'
Adaptive evidence	Expected information gain over missing or stale modalities	Minimum next optional check
Unit forecast	Apply models to planned roster/deployment for next 14 days, aggregate with DP	Welfare Weather Forecast per unit
Anti-gaming checks	PVT effort validity (too fast/too uniform), straight-lining detection, self vs objective discordance	Quality flag, not a penalty

8.3 Confidence logic

Confidence is separate from concern. It combines calibration, evidence coverage, modality quality, recency and inter-domain agreement. High operational load with no fresh individual evidence yields **low confidence plus an optional check request**, not an alert. Discordance is handled respectfully: poor self-report with normal objective signals always yields a **confidential support offer**, never dismissal.

8.4 Synthetic data generator (answers 'is the model just learning your generator?')

- 300 personnel, 3 battalions, 180 days; each person has a **hidden latent strain state** driven by operational load with individual sensitivity (random effects).
- Observed signals are noisy, delayed functions of latent strain, with missingness, device noise and 10% deliberately inconsistent self-reports.
- Confounders included: seasonal duty spikes, individuals with naturally slow reaction times, resilient high-load people.
- Labels (deterioration events) come from the latent state with a threshold plus noise. The model never sees the latent variable or generator rules.

- Baselines compared: single-threshold rule on duty hours, and a population-only model. SAHARA must beat both on lead time and PR-AUC.
- Everything is labelled SIMULATED in UI and metrics. Real validation plan in Section 23.

9. Crisis Safe-Route (Safety Net)

Non-negotiable design rule

SAHARA does not predict or score suicide risk. But if any crisis signal appears, the system stops analysing and connects a human immediately.

Trigger	Immediate response on phone	Human response
User taps 'I need help now'	Full-screen: call Tele-MANAS 14416, call unit counsellor, call buddy. Calming breathing guide.	Counsellor + welfare officer alerted on crisis lane; callback SLA 30 minutes
WHO-5 score very low (e.g., raw score ≤ 7) or 'urgent support' item answered Yes	Gentle message + same options; no score shown	Welfare officer alerted within 2 hours; mandatory human contact
Crisis keyword in optional free-text (on-device keyword match only)	Same safe-route; text itself never sent to server	Flag only (no content) to counsellor

All SLA values are defaults, configurable to force policy. Crisis events bypass commander views entirely and are visible only to the counsellor and welfare officer on duty, with break-glass audit.

10. Automated Alert Engine

Flow

Risk event → Policy rules + confidence gate → Route by tier → Notify (push / SMS / email) → Acknowledge within SLA → Auto-escalate if missed

Tier	Condition	Recipient	Ack SLA	Escalation
T0 Crisis	Crisis Safe-Route trigger	Duty counsellor + welfare officer	30 min	Senior welfare officer / Medical Officer
T1 High	Any dimension High with confidence ≥ 0.7 sustained 3+ days	Unit welfare officer	24 h	Senior welfare officer at 48 h
T2 Moderate	Moderate with evidence agreement ≥ 2 domains	Unit welfare officer (digest)	72 h	Re-queue as T1 if worsening
T3 Evidence request	Low confidence with rising operational load	Personnel (optional check request)	N/A	None, never escalates to staff
Unit hotspot	Unit forecast High for $\geq 20\%$ of strength (DP aggregate)	Commander + planner	7 days	Formation welfare cell

Alert fatigue controls: deduplication, daily digest for T2, capacity-aware ranking (recall at review capacity), and snooze with mandatory reason.

11. Welfare Intervention Recommendation System

Flow

Concern surfaced → Welfare review → Recommended options → Human selects → Roster / counselling / welfare action → Follow-up → Measure recovery

Situation	Recommended options (ranked)	Follow-up
Burnout High, workload driven	Roster rebalancing via optimizer, recovery window, defer non-critical training, welfare conversation	7 days
Acute Stress High after incident exposure	Critical incident debrief, counsellor session, short rest period	3 days
Emotional Fatigue with family separation / recent transfer	Family connect (video call slot), leave planning, welfare need follow-up	7-14 days
Welfare Concern open > 14 days	Escalate grievance to responsible branch, update personnel with status	Until closed
Discordance (feels bad, signals normal)	Confidential support offer, counsellor option, no adverse inference	Officer decides
Low confidence	Optional readiness check only	7-14 days

11.1 Roster optimizer (workload balancing)

Google OR-Tools CP-SAT. Inputs: required posts, shift coverage, skills, policy rest limits, current assignment, approved non-availability, fairness (spread of night duties). Objective: minimize total predicted strain and inequality while keeping all hard constraints. Output: up to 3 feasible plans with coverage check, fairness index and projected risk change. **Advisory only; command authority approves.**

11.2 Intervention effectiveness tracker

Each intervention records pre/post trajectory. Aggregated (never per-person to commanders) to show which interventions work for which duty types. Roadmap: contextual bandit to personalize recommendations after governance approval.

12. Welfare Need Module and Resource Planner

12.1 Welfare Need and Grievance module

Categories: family medical emergency, housing/quarters, children education, financial stress (loans, pay anomaly), leave request, transfer request, pending allowances. Each need has status, owner branch, SLA and closure feedback. Unresolved needs feed the Welfare Concern dimension. This addresses root causes, not just symptoms.

12.2 Welfare Resource Planner (commander and HQ view)

- Forecast counsellor sessions needed per battalion for the next 30 days.
- Heat map of welfare need categories by sector (DP protected) for budget and resource allocation.
- Aggregate retention risk trend (burnout + welfare concern + leave deprivation) for HR planning, never individual.
- Workload Balance Index per unit: distribution of night duty and deployment burden.

13. Technical Architecture

Flow

Mobile App / Kiosk + HRMS Adapter → API Gateway + Privacy Gateway → Encrypted Postgres + Feature Store → ML Engine + OR-Tools + Alert Engine → Welfare / Commander / Admin apps

13.1 Stack

Layer	Prototype choice	Responsibility
Mobile	React + Vite PWA (Capacitor wrap for Android APK), i18next, IndexedDB offline queue	Consent, check-in, readiness test, PPG, My Trend, My Data Mirror
Web dashboards	React + Tailwind + Recharts	Welfare console, commander, planner, admin
API	FastAPI (Python), OpenAPI, Pydantic, JWT/OIDC	Auth, RBAC/ABAC, workflows, validation
Data	PostgreSQL with row level security; Redis for jobs/alerts	Cases, features, audit, consent
ML	pandas, scikit-learn, XGBoost, SHAP, ruptures (change-points), MLflow	Baselines, 4 models, calibration, explanation
Optimization	Google OR-Tools CP-SAT	Roster alternatives
Alerts	Background worker + Web Push / SMS gateway stub / email	Tiered notifications, SLA timers
Deploy	Docker Compose (prototype); Kubernetes on private cloud (production)	One-command demo
Optional assistant	On-prem small LLM + RAG over approved welfare SOPs (Tier 3)	Policy lookup only, never risk scoring

13.2 Data flow

Stage	Processing	Output
Collection	Client validation, consent check, on-device feature computation, raw discard	Encrypted event with pseudonymous ID
Privacy Gateway	Tokenization, minimization, purpose tag, retention tag, authorization	Authorized feature payload
Feature layer	Rolling windows, baseline deviation, change-points, missingness	Versioned feature snapshot
Inference	4 risk models, calibration, confidence, SHAP, counterfactuals	RiskAssessment record
Alert engine	Tier rules, dedup, SLA timers	Notifications + queue
Action	Human decision, intervention, roster scenario, follow-up	Audit event + recovery outcome

13.3 HRMS integration

Adapter pattern: REST pull/push, CSV batch or SFTP drop from existing force HRMS. Canonical event schema (duty, leave, deployment, transfer, training, incident). mTLS service identity, unit-scoped tokens, schema validation, idempotency keys, replay protection. Only fields needed for features are ingested (field allow-list).

14. Privacy, Security and Ethics Framework

14.1 Privacy controls

Control	Implementation	Demo proof
Consent manager	Per-modality, versioned, withdraw anytime, consent receipts	Consent screen + receipt
Purpose binding	Every record tagged 'welfare'; policy engine denies non-welfare purposes	Simulated external HR/disciplinary client requesting welfare data returns 403
Minimization	Derived features only; no raw frames, audio, text, GPS	Inspect DB payload
Pseudonymization	Separate identity vault; rotating pseudonyms in analytics	Queue shows P-IDs
Anonymization	Minimum cell size $k \geq 10$ (small groups suppressed) + Laplace differential privacy noise on aggregates with a privacy budget	Small unit shows 'insufficient group'
Identity reveal	Break-glass: reason, second approver, time-limited, logged, visible in My Data Mirror	Simulated reveal
Retention	Purpose-specific retention; auto deletion jobs	Retention tag on events
Transparency	My Data Mirror + model cards + plain-language notice	Personnel access log

14.2 DPDP Act 2023 alignment

DPDP principle	SAHARA mechanism
Notice and consent	Plain-language notice in user's language; granular, itemized consent; consent receipts
Purpose limitation	Purpose-binding policy engine; welfare-only processing
Data minimization and storage limitation	Derived features, retention schedules, auto deletion
Rights of data principal	Access, correction, erasure and grievance through My Data Mirror
Reasonable security safeguards	Encryption, RBAC/ABAC, audit, SIEM, breach response
Breach notification	Incident runbook: report to authority and CERT-In within prescribed timelines

Note: where processing is carried out by the State for security or other exempted purposes, SAHARA still applies these principles voluntarily to build trust.

14.3 Security controls

Area	Controls
Identity	OIDC, MFA for staff, device binding for personnel, short-lived tokens
Transport and storage	TLS 1.3 / mTLS between services; AES-256 at rest; keys in HSM/KMS separate from data
Application	OWASP ASVS Level 2 controls, input validation, rate limits, CSP headers
Audit	Append-only, hash-chained audit log (tamper evident); SIEM export
Mobile	Encrypted local store, root detection, screenshot block, remote wipe
Operations	Air-gapped / private network deployment, least privilege, separation of duties, backup and DR

14.4 Threat model (STRIDE summary)

Threat	Example	Mitigation
Spoofing	Stolen staff credentials	MFA, device binding, anomaly login alerts
Tampering	Editing case history	Hash-chained audit, RLS, immutable events
Repudiation	Officer denies viewing data	Signed audit entries + My Data Mirror
Information disclosure	Commander infers individual from small unit	k>=10 suppression + differential privacy + query budget
Denial of service	Flood API	Rate limiting, queueing, private network
Elevation of privilege	Commander calls welfare endpoint	ABAC with purpose + unit claims, 403 + alert

14.5 Responsible AI

- Model cards for every model: intended use, limits, training data (synthetic), metrics, fairness results.
- Fairness audit across rank, duty type, posting tier and gender (evaluation only, never as decision input).
- Ethics and Welfare Board sign-off before any production model release; drift monitoring and quarterly review.
- Welfare officer feedback ('helpful / not helpful / false alarm') captured for governed retraining.

15. Built for Indian Forces

Need	SAHARA response
Linguistic diversity	English + Hindi at launch, regional languages via i18n packs; icon-first check-in
Remote posts, weak connectivity	Offline-first app with encrypted sync queue; low-bandwidth mode; kiosk tablet at post
Personal phone restrictions in ops areas	Shared kiosk with per-person PIN; periodic check-in at base
National mental health ecosystem	Referral route to Tele-MANAS (14416) and force counselling cells
Family separation	Family connect scheduling, leave planning assistant, family welfare needs
Data sovereignty	Fully on-prem; open-source stack; no foreign cloud dependency; indigenous IP
Hierarchy and stigma	Commander never sees individuals; welfare-only purpose binding; anonymous support option

16. Data Model and APIs

16.1 Core entities

Entity	Key fields
PersonnelProfile	person_id, pseudonym, unit_id, rank_band, duty_type, consent_state, baseline_status, language
ConsentRecord	consent_id, person_id, modality, purpose, version, status, receipt_hash, timestamp
OperationalEvent	event_id, person_id, type (duty/leave/deployment/transfer/training/incident), start, end, intensity, source
ReadinessAssessment	assessment_id, person_id, modality, metrics JSON, quality, device_ts

Entity	Key fields
CheckIn	checkin_id, person_id, energy, sleep, workload, wellbeing, burnout_pulse, fatigue, ts
FeatureSnapshot	snapshot_id, person_id, window, features JSON, feature_version
RiskAssessment	id, person_id, model_version, dimension, band, probability, confidence, coverage, factors JSON, counterfactuals JSON
Alert	alert_id, tier, person_id or unit_id, reason, status, ack_by, sla_due, escalated_to
WelfareCase	case_id, person_id, state, owner, priority, followup_at
Intervention	id, case_id, type, rationale, approved_by, start, end, pre_score, post_score
WelfareNeed	need_id, person_id, category, status, owner_branch, sla_due, closure_feedback
RosterScenario	scenario_id, unit_id, constraints, changes, feasible, fairness_index, strain_delta
UnitForecast	unit_id, date, dimension, dp_value, epsilon_used, suppressed
AuditEvent	event_id, actor, role, action, resource, purpose, ts, prev_hash, hash

16.2 API surface

Endpoint	Purpose	Authorization
POST /v1/consents	Grant or withdraw modality consent	Personnel (self)
POST /v1/checkins	Submit check-in / pulse	Personnel (self)
POST /v1/readiness/summary	Submit on-device derived features	Personnel, consented modality
POST /v1/crisis	Trigger Crisis Safe-Route	Personnel (self)
POST /v1/welfare-needs	Raise welfare need	Personnel (self)
GET /v1/me/trend GET /v1/me/access-log	My Trend My Data Mirror	Personnel (self)
POST /v1/integrations/hrms/events	Ingest HRMS events	Service identity, unit scoped
GET /v1/welfare/queue	Prioritized alerts	Welfare officer, unit + purpose
GET /v1/welfare/cases/{id}	Evidence, factors, counterfactuals	Case owner
POST /v1/welfare/cases/{id}/actions	Record action and follow-up	Welfare role, reason required
POST /v1/alerts/{id}/ack	Acknowledge alert	Recipient
GET /v1/commander/units/{id}/aggregate	DP aggregates + forecast	Commander, aggregate only
POST /v1/roster/scenarios	Run optimizer	Planner / commander
GET /v1/planner/resources	Counsellor demand, need heat map	HQ welfare cell
GET /v1/audit	Search audit	Auditor

17. Functional Requirements

ID	Requirement	Priority	Acceptance signal
FR-01	Granular consent and check-in under 20 seconds	Must	Consent receipt persists; withdrawal stops collection

ID	Requirement	Priority	Acceptance signal
FR-02	Ingest HRMS events incl. transfers and leave; compute 7/30/90-day features	Must	Feature panel shows all 6 PS indicators
FR-03	Readiness test and camera PPG computed on device; summaries only sent	Must	No raw tap/frame data server side
FR-04	Personal baseline with cold-start shrinkage; change-point marking	Must	Chart shows baseline band and shift date
FR-05	Four risk dimensions with band, probability, confidence, factors	Must	Reproducible outputs on seeded data
FR-06	Counterfactual 'what would help' on actionable features	Must	At least 1 actionable counterfactual per High alert
FR-07	Low confidence triggers minimum optional evidence request	Must	Confidence 40% to 97% after check (measured in prototype)
FR-08	Tiered alert engine with SLA and escalation	Must	Unacknowledged T1 escalates in demo (time-compressed)
FR-09	Crisis Safe-Route with Tele-MANAS and counsellor routing	Must	Crisis tap creates T0 alert; no score shown
FR-10	Welfare case workflow with follow-up and recovery tracking	Must	Case action in audit; post-score shown
FR-11	Commander DP aggregates, 14-day unit forecast, no individual access	Must	Individual endpoint returns 403 and is logged
FR-12	Roster optimizer with hard constraints and fairness	Must	Feasible plan with projected strain drop
FR-13	My Data Mirror access log for personnel	Must	Officer view appears in personnel timeline
FR-14	Welfare Need module with SLA	Must	Need raised, routed, closed
FR-15	Welfare Resource Planner	Should	Counsellor demand forecast chart
FR-16	Hindi language and offline check-in sync	Must	Toggle language; airplane mode check-in syncs
FR-17	Buddy nudge, anonymous counsellor chat, wearable connector, kiosk mode	Could	Shown if time permits
FR-18	Hash-chained audit and purpose-binding enforcement	Must	Tamper test breaks chain verification

18. Evaluation Metrics

Dimension	Metric	Target (simulated)
Early detection	Onsets detected and warning time vs a duty-load rule at the SAME alert budget	Report honestly. Prototype: both warn early; SAHARA gains on precision (burnout PR-AUC 0.60 vs 0.46 HRMS-only)
Model quality	PR-AUC, AUROC, recall at review capacity, Brier, ECE per dimension	Beat both baselines; ECE < 0.05
False alerts	Alerts per 100 personnel per week; precision at T1	< 3 per 100; precision > 0.6
Explainability	Alerts with 3 factors + counterfactual + recency	100%
Fairness	Recall and false-alert gap across rank/duty/posting cohorts	Gap < 5 points; reported openly

Dimension	Metric	Target (simulated)
Usability	Check-in time, SUS score, task success	< 20 s; SUS > 75; > 90%
Privacy/security	Forbidden requests denied and logged; raw data absence	100%
Optimization	Hard constraint satisfaction; strain reduction; fairness index	100%; measurable reduction
Alert operations	T1 acknowledged within SLA	> 95% in simulation
Recovery loop	Follow-up completion; post-intervention trend	Shown per case

19. Build Plan for SIH Grand Finale

19.1 Team of 6

Role	Ownership	Finale outputs
M1 Team Lead / Product	PRD, PS traceability, pitch, judge Q&A	Deck, demo script, risk register
M2 Mobile / Frontend	SAHARA Mobile (check-in, readiness test, PPG, My Trend, My Data Mirror, crisis)	Working mobile flows, Hindi toggle
M3 Dashboard / Frontend	Welfare console, commander, planner, admin/audit	All role views
M4 Backend / Security	FastAPI, auth, RBAC/ABAC, privacy gateway, alert engine, audit chain	OpenAPI, role tests, 403 proof
M5 ML Engineer	Generator, features, baselines, 4 models, SHAP, counterfactuals, forecast	Metrics notebook, model cards
M6 Optimization / QA / DevOps	OR-Tools roster, Docker Compose, seeded demo, tests, fallback video	One-command demo, runbook

19.2 Before the finale (pre-build)

Compress or stretch these weeks to the actual time left before the Grand Finale; the order of work stays the same.

Week	Deliverable	Exit criterion
1	Synthetic generator, schema, wireframes, threat model	Seeded DB with 300 personnel x 180 days
2	Backend core + auth/RBAC + mobile check-in and readiness test	End-to-end ingestion works
3	Baselines, 4 models, calibration, SHAP, confidence	P-104 scenario produces deterministic output
4	Welfare console, alert engine, crisis route, case workflow, welfare need module	Role separation demonstrated
5	Commander DP view, forecast, roster optimizer, My Data Mirror	Feasible plan + DP proof
6	Hindi, offline sync, metrics page, polish, fallback video	7-minute demo rehearsed 10 times

19.3 36-hour finale plan

Hours	Focus	Judge round readiness
0-4	Environment up, seeded data verified, fix anything broken since rehearsal, incorporate mentor feedback	Round 1: pitch + traceability matrix + live core loop

Hours	Focus	Judge round readiness
4-14	Tier 2 features: resource planner, buddy nudge, anonymous chat, wearable stub	Show progress vs Round 1 feedback
14-20	Hardening: security tests, fairness report, edge cases, crisis flow polish	Round 2: deep dive on AI + privacy
20-30	UX polish, multilingual, kiosk mode, performance, final metrics	Round 3 prep
30-36	Freeze, rehearsal, backup video, deck final	Final evaluation: 7-minute demo + Q&A

20. Finale Demo Script (7 minutes)

Scenario: P-104, 3rd Battalion. 6 night duties in 10 days and 4 more planned next week, 31-day continuous deployment, leave cancelled, transferred 30 days ago to a post 1,450 km from home, open leave request for a family surgery. HRMS load rises but confidence is only 40% because individual evidence is stale (last check-in 9 days ago, last readiness test 27 days ago).

Time	What judges see	Proof point
0:00-0:40	Problem, 'welfare not surveillance' promise, traceability matrix	Every PS line covered
0:40-1:40	P-104 gives consent, 10-sec check-in in Hindi, 90-sec readiness test	Low burden, voluntary, multilingual
1:40-2:15	Privacy Gateway payload: pseudonymous derived features only	No raw data, no public LLM
2:15-3:15	Baseline chart + change-point; confidence rises 40% to 97%; all four evidence domains agree; T1 alert reaches welfare officer	Personal baseline + adaptive evidence
3:15-4:00	T1 alert pushes to welfare officer phone; SHAP factors + counterfactual 'what would help'	Automated, explainable, actionable
4:00-4:50	Commander sees DP unit forecast hotspot; tries individual view: 403 logged; planner runs optimizer	Workload balancing with no disclosure
4:50-5:30	Welfare officer schedules counselling + resolves leave need; follow-up shows recovery	Closed loop
5:30-6:10	P-104 opens My Data Mirror: sees officer's access and purpose. Crisis button demo (T0 lane).	Trust + safety
6:10-7:00	Metrics page (lead time, PR-AUC, fairness), architecture, validation roadmap	Honest, deployable

21. Judge Q&A Defence Sheet

Question	Answer
Is this surveillance of soldiers?	No. Voluntary modalities, no GPS/mic/messages, welfare-only purpose binding, commanders see only DP aggregates, and personnel see every access in My Data Mirror.
What is your accuracy?	On simulated data we report PR-AUC, calibration and lead time vs baselines. We do not claim clinical accuracy; real validation needs a governed study (Section 23).
Where do labels come from in reality?	Welfare officer case outcomes, validated periodic wellbeing scores and counsellor-confirmed support needs, collected under ethics approval.

Question	Answer
What if someone is suicidal?	Crisis Safe-Route: no scoring, immediate human contact, Tele-MANAS 14416, counsellor with 30-min SLA.
What about false positives and stigma?	Confidence gating, multi-domain agreement, pseudonymous queue, supportive language, human review, no adverse action allowed.
What if personnel lie on self-reports?	We do not rely on one source. Objective and self signals are fused; discordance leads to a confidential support offer, not a penalty. Anti-gaming quality flags lower confidence.
Why not use ChatGPT/LLM?	Personnel data cannot go to public APIs. Prediction uses explainable ML on-prem. An optional local LLM only answers welfare policy questions.
New recruit has no history?	Cold-start shrinkage to cohort baseline, shifting to personal baseline as data grows.
Can a commander misuse the data?	ABAC with purpose, $k \geq 10$ + differential privacy, 403 + alert on violation, audit visible to personnel.
No smartphone at forward posts?	Kiosk mode with PIN, offline sync, and HRMS-only risk with low confidence flag.
How is it integrated with existing HRMS?	Adapter pattern (REST/CSV/SFTP), canonical schema, field allow-list, mTLS.
Is it legal under DPDP?	Consent, purpose limitation, minimization, rights portal, breach runbook mapped in Section 14.2.
How does it reduce workload?	Roster optimizer produces feasible plans under coverage, skill and rest constraints with fairness; commander approves.
How is it different from other teams?	Personal baseline + adaptive evidence, 4 calibrated dimensions, counterfactual actions, crisis route, My Data Mirror, DP commander view, closed-loop recovery.
What will it cost to deploy?	Open-source stack, no licence fees. Models are lightweight tree models: nightly inference for 1 lakh personnel runs on standard CPU servers, no GPU. Main cost is existing force data-centre capacity and training of welfare staff.
What if HRMS data is incomplete or not digitized?	Adapter accepts CSV batch uploads; missing domains lower confidence rather than breaking the model; self-report and readiness evidence still work.
Who is accountable if the AI misses someone?	The AI only prioritizes; welfare officers keep all existing channels. Every decision is human-owned and audited. Metrics track recall and missed cases for review.
How will it scale?	Stateless API, per-unit partitioning, batch nightly inference + event-driven updates; tested on 300 synthetic, designed for 1 lakh+ personnel per force.

22. Market, Scalability and Adoption

Segment	Fit	Adaptation
CAPFs (CRPF, BSF, CISF, ITBP, SSB, AR, NSG)	Primary target	Unit/battalion hierarchy config
Indian Armed Forces	High	Air-gapped deployment, service-specific duty types
State Police	High (shift-heavy)	State HRMS adapters, regional languages
NDRF / SDRF / Fire / Emergency services	High (trauma exposure)	Incident exposure weighting
High-stress Government workforces (Railways, health workers)	Medium	Duty type templates
Corporate HR wellness	Medium	SaaS tenant, HRIS connectors
International security forces	Long term	Localization, compliance packs

Adoption path: pilot in one battalion (6 months) with ethics approval, then formation, then force-wide. Multi-tenant architecture with configurable policies per force. Open-source core with indigenous IP; cost dominated by on-prem hardware already available in force data centres.

23. Risks, Mitigations and Validation Roadmap

Risk	Mitigation
False positives and stigma	Confidence gating, human review, supportive language, no punitive use
Weak real-world labels	Synthetic demo; governed pilot with DIPR / force psychologists for validation
Data exposure	On-prem, minimization, encryption, DP, audit
Low engagement	20-second interaction, personal benefit, adaptive requests only when useful
Alert fatigue	Tiering, digest, dedup, capacity-aware ranking
Overbuilding at finale	Tier discipline: Tier 1 demo-grade, Tier 2 show, Tier 3 slides
Roster conflicts with mission	Hard constraints, advisory only, command approval

23.1 Post-SIH roadmap (Tier 3)

- Governed 6-month pilot with ethics approval and data protection impact assessment.
- Federated learning across units so raw data never centralizes.
- On-prem multilingual LLM assistant for welfare policy and family scheme lookup.
- Contextual bandits to personalize interventions based on measured effectiveness.
- Integration with force hospitals and counselling cells for referral workflows.

24. Release Acceptance Criteria

Category	Criterion
End to end	P-104 flows from HRMS event to alert, review, intervention, follow-up and recovery with no manual DB edits
PS coverage	Every row of Section 2 demonstrable live or on metrics page
4 risk dimensions	Each shows band, probability, confidence, top 3 factors and caution text
Adaptive confidence	Visible 40% to 97% confidence change after optional check
Alerts	T1 push notification + simulated SLA escalation + T0 crisis lane
Role separation	Personnel, welfare, counsellor, commander, planner, admin logins expose different data; 403 logged
Privacy proof	DB contains derived metrics only; DP aggregate with suppression; My Data Mirror entry
Roster	At least one feasible plan passes coverage, skill, rest constraints with projected strain drop
Metrics	Metrics page: lead time, PR-AUC, calibration, fairness, security tests, limitations
Resilience	Seeded offline data, recorded fallback video, operator runbook

Final recommendation

Build Tier 1 to demo-grade perfection, show 2-3 Tier 2 features, keep Tier 3 on slides. A credible, safe, explainable closed loop that ticks every PS line will beat a broad but shallow 'AI stress detector'.

